Concept 5 | Advanced Theorems and Applications

English Student Edition • Focused formulas and guided practice

Formula opener

Addition

$$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$$

Double angle

$$\sin 2A = 2 \sin A \cos A$$

Cosine

$$\cos 2A = \cos^2 A - \sin^2 A$$

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Classified Practice **Advanced Theorems and Applications**

Q001 **Written**

Find the maximum and minimum of $y = sin^2\theta + cos\theta + 1, 0le\theta < 2pi$, and the corresponding θ.

Q002 **Written**

Find the maximum and minimum of $y = cos^2\theta - sin\theta, 0le\theta < 2pi$, and the corresponding θ.

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Q003 **Written**

Solve the four given maximum/minimum problems on $0le\theta < 2pi$**, including** $y = \sin^2\theta + \sin\theta - 1$**,** $y = \cos^2\theta - \cos\theta + 1$**,** $y = 2\sin\theta + \cos^2\theta - 1$**, and** $y = -2\sin^2\theta - 2\cos\theta + 1$**. Include the given challenge** $y = \cos^2\theta - \sin\theta$**.**

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Q004 **Written**

Use addition theorems to find $\sin 75°$ **and** $\cos(\pi/12)$**.**

Q005 **Written**

Given $\sin\alpha = \frac{\sqrt{3}}{2}$ **in QII and** $\cos\beta = -\frac{3}{5}$ **in QIII, find** $\sin(\alpha+\beta)$ **and** $\cos(\alpha-\beta)$**.**

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Q006 **Written**

Complete the given addition-theorem exercises $exactvalues, quadrant - controlled\alpha, \beta, and the three - angle challenge(\sin\alpha = 0.6, \cos\beta = 0.8, \tan\gamma = 0.75)$.

Q007 **Written**

Find $\tan 105°$**, then find the angle between** $y = 2x + 1$ **and** $y = -3x + 2$**.**

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Q008 Written

Find the angle between each given pair of straight lines, including $y = \frac{1}{3}x$ with $y = 2x$, and $8x+2y-3 = 0$ with $-5x+3y+1 = 0$.

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Q009 Written

Find the exact angles for all given line pairs and solve the challenge: a line through $P(2,3)$ makes acute angle $\tan\theta = 3/4$ with $x + 2y - 4 = 0$; give both line equations and the areas of the triangles with the y-axis.

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Q010 **Written**

Given $\sin\alpha = 1/3$ **with α in QII, find** $\sin 2\alpha, \cos 2\alpha, \tan 2\alpha$**.**

Q011 **Written**

Given $\sin\alpha = -2/3$ **with** $2\pi < \alpha < 3\pi$, **find** $\sin 2\alpha, \cos 2\alpha, \tan 2\alpha$.

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Q012 **Written**

Solve the given double-angle exercises for given sine/cosine values and the challenge $\cos\theta = 0.6, \pi < \theta < 2\pi$**, evaluating** $\cos 3\theta \tan 2\theta$**.**

Q013 **Written**

Given $\cos\alpha = 3/5$, $0 < \alpha < \pi$, **find** $\sin(\alpha/2)$, $\cos(\alpha/2)$, $\tan(\alpha/2)$, **and evaluate** $\cos(5\pi/8)$.

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Q014 **Written**

Given $\cos\alpha = -1/3$, $0 < \alpha < \pi$, **find** $\sin(\alpha/2), \cos(\alpha/2), \tan(\alpha/2)$, **and evaluate** $\sin(3\pi/8)$.

Q015 **Written**

Solve the given half-angle exercises for the three quadrant intervals and the challenge with $\tan A = \frac{\sqrt{5}}{2}, 4\pi < A < 5\pi$, $\cos B = -\frac{\sqrt{5}}{3}, 5\pi < B < 6\pi$.

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Q016 **Written**

Solve on $0 \leq \theta < 2\pi$: $a \sin 2\theta + \sin\theta = 0$, $b \cos 2\theta - 7\cos\theta + 4 \geq 0$.

Q017 **Written**

Solve the given equation $\sin 2\theta + \cos\theta = 0$ **on** $0 \leq \theta < 2\pi$**.**

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Q018 Written

Solve all given double-angle equations and inequalities on $0 \le \theta < 2\pi$, including the challenge $\cos 2\theta - 7\cos\theta + 4 \ge 0$.

Q019 **Written**

Rewrite $\sin\theta - \sqrt{3}\cos\theta$ **as** $r\sin(\theta+\alpha)$, $r > 0, -\pi < \alpha < \pi$.

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Q020 **Written**

Rewrite the given expressions $\sin\theta - \cos\theta$, $\sin\theta + \cos\theta$, $3\sin\theta - \sqrt{3}\cos\theta$, **and** $3\sin\theta - \cos\theta$ **in harmonic-addition form.**

Q021 **Written**

Rewrite all given harmonic-addition exercises in the form $r\sin(\theta+\alpha)$**, including** $2\sin\theta + 2\cos\theta$ **and the challenge** $\sin\theta - \sqrt{3}\cos\theta$**.**

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Q022 **Written**

Solve on $0 \leq \theta < 2\pi$: $a \sin\theta - \cos\theta = 1$, $b \sin\theta + \cos\theta < 3/2$.

Q023 **Written**

Solve $\sin\theta \leq \cos\theta$ **on** $0 \leq \theta < 2\pi$.

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Q024 **Written**

Solve all given harmonic-addition equations and inequalities on $0 \le \theta < 2\pi$, including $\sin\theta + \cos\theta + 1 = 0$ and $\sin\theta - \cos\theta + 3/2 > 0$.

Q025 **Written**

Find the maximum and minimum of $y = 2\sin(\theta - \pi/3)$**, and of** $y = 2\cos 2\theta + 4\cos\theta + 1$, $0 \le \theta < 2\pi$**.**

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Q026 Written

Find the maximum and minimum of the given functions $y = -\frac{4}{3}\sin^2\theta + \cos\theta$ and $y = \cos 2\theta - 2\cos\theta$.

Q027 Written

Solve all given maximum/minimum problems in the final lesson, including absolute-value and double-angle forms, and state the corresponding angles where requested.

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Q028 MCQ

The substitution for $\sin^2\theta + \cos\theta + 1$ should be:

Choose the correct option.

A $t = sin$ theta

B $t = cos$ theta

C $t = tan$ theta

D $t = theta$

Q029 MCQ

The maximum of $\sin^2\theta + \cos\theta + 1$ **is:**

Choose the correct option.

A 0

B 1

C 9/4

D 4

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Q030 MCQ

An extremal vertex is valid only if t lies in:

Choose the correct option.

A [-2,2]

B [-1,1]

C [0,1]

D all real numbers

Q031 **MCQ**

The identity for sin(a+b) is:

Choose the correct option.

A sin alpha cos beta + cos alpha sin beta

B sin alpha sin beta

C cos alpha cos beta

D tan alpha+tan beta

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Q032 **MCQ**

sin 75 degrees equals:

Choose the correct option.

A (sqrt6-sqrt2)/4

B (sqrt6+sqrt2)/4

C 1/2

D sqrt3/2

Q033 **MCQ**

A missing trig value is selected using:

Choose the correct option.

A the quadrant
B the denominator
C the period only
D a graph scale

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Q034 MCQ

The slope of y=mx+k is:

Choose the correct option.

A k
B m
C 1/m
D m+k

Q035 MCQ

The angle formula uses denominator:

Choose the correct option.

A 1+m1m2

B m1-m2

C m1+m2

D m1m2

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Q036 MCQ

For an acute angle, if tan theta1 is negative we use:

Choose the correct option.

A theta1

B pi-theta1

C 2theta1

D -theta1 only

Q037 **MCQ**

sin(2a) equals:

Choose the correct option.

A sin alpha

B 2 sin alpha cos alpha

C cos squared alpha

D tan alpha

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Q038 MCQ

cos(2a) can be written as:

Choose the correct option.

A cos squared alpha - sin squared alpha

B 2sin alpha

C tan alpha

D 1+sin alpha

Q039 **MCQ**

For theta in QIV, sin theta is:

Choose the correct option.

A positive

B negative

C zero always

D undefined

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Q040 MCQ

The half-angle formula for sin(a/2) uses:

Choose the correct option.

A (1-cos alpha)/2

B (1+cos alpha)/2

C tan alpha

D cos 2alpha

Q041 **MCQ**

The sign of a half-angle root is chosen from:

Choose the correct option.

A the original quadrant interval

B the coefficient only

C the denominator

D the period of tan

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Q042 MCQ

If cos a=-1/3 and 0<a<pi, cos(a/2) is:

Choose the correct option.

A negative
B positive
C zero
D undefined

Q043 **MCQ**

For double-angle equations, first rewrite in:

Choose the correct option.

A one trig function

B two unrelated angles

C degrees only

D a logarithm

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Q044 MCQ

The factorisation of sin(2 theta)+sin theta is:

Choose the correct option.

A sin theta(2cos theta+1)

B cos theta(2sin theta-1)

C sin theta+cos theta

D 2sin theta

Q045 **MCQ**

A tangent asymptote is always:

Choose the correct option.

A included
B excluded
C a maximum
D an endpoint

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Advanced Theorems and Applications

Q046 MCQ

For a sin theta+b cos theta, r is:

Choose the correct option.

A a+b

B sqrt(a^2+b^2)

C a-b

D ab

Q047 MCQ

The harmonic target form is:

Choose the correct option.

A r sin(theta+alpha)

B r cos theta

C a tan theta

D sin theta only

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Q048 MCQ

The angle a is located from point:

Choose the correct option.

A P(a,b)
B P(r,theta)
C the origin only
D the x intercept

Q049 **MCQ**

After synthesis, t is usually:

Choose the correct option.

A theta+alpha

B $theta^2$

C tan theta

D cos theta

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Q050 MCQ

If max(sin theta+cos theta)=sqrt2, then sin theta+cos theta<3/2 is:

Choose the correct option.

A never true
B always true
C true only at pi
D undefined

Q051 **MCQ**

In a strict inequality, equality endpoints are:

Choose the correct option.

A included

B excluded

C doubled

D random

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Q052 MCQ

The range of 2sin(theta-pi/3) is:

Choose the correct option.

A [-1,1]
B [-2,2]
C [0,2]
D [-3,3]

Q053 **MCQ**

For a double-angle maximum problem, use:

Choose the correct option.

A the double-angle formula then a bounded variable

B a logarithm

C only a graph

D no substitution

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Q054 MCQ

A vertex outside [-1,1] is:

Choose the correct option.

A an allowed extremum

B rejected

C always the maximum

D always the minimum

Quiz MCQ 1 **Concept Quiz · MCQ**

The maximum of $2\sin(\theta - \pi/3)$ is:

Choose the correct option.

A -2

B -1

C 1

D 2

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Quiz MCQ 2 **Concept Quiz · MCQ**

The addition-theorem value $\sin 75°$ is:

Choose the correct option.

A $(\sqrt{6}+\sqrt{2})/4$
B $(\sqrt{6}-\sqrt{2})/4$
C $1/2$
D 1

Quiz MCQ 3 **Concept Quiz · MCQ**

For $y = 2x^2 + 4x + 1$, the vertex x-value is:

Choose the correct option.

A -2
B -1
C 1
D 2

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Quiz MCQ 4 **Concept Quiz · MCQ**

The angle between slopes 2 and -3 is:

Choose the correct option.

A $\pi/6$
B $\pi/4$
C $\pi/3$
D $\pi/2$

Quiz WRITTEN 5 **Concept Quiz · Written**

Find the maximum and minimum of $y = \sin^2\theta + \cos\theta + 1$ on $0 \le \theta < 2\pi$.

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Quiz WRITTEN 6 **Concept Quiz · Written**

Find $\sin 2\alpha, \cos 2\alpha, \tan 2\alpha$ **when** $\sin \alpha = 1/3$ **and α is in QII.**

Quiz WRITTEN 7 **Concept Quiz · Written**

Rewrite $3\sin\theta - \sqrt{3}\cos\theta$ **as** $r\sin(\theta + \alpha)$**.**

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Quiz WRITTEN 8 **Concept Quiz · Written**

Solve $\sin\theta \leq \cos\theta$ **on** $0 \leq \theta < 2\pi$.

Answer key

Use the question number shown on each page.

Q001 $\max y = \frac{9}{4}$ at $\theta = \frac{\pi}{3}, \frac{5\pi}{3}$; $\min y = 0$ at $\theta = \pi$

Q002 $\max y = \frac{5}{4}$ at $\theta = \frac{7\pi}{6}, \frac{11\pi}{6}$; $\min y = -1$ at $\theta = \frac{\pi}{2}$

Q003 See the numbered solution: substitute $t = \sin\theta$ or $t = \cos\theta$, complete the square, then test $t = \pm 1$.

Q004 $\sin 75° = \cos \pi/12 = \frac{\sqrt{6}+\sqrt{2}}{4}$

Q005 $\sin \alpha + \beta = \frac{4-3\sqrt{3}}{10}$; $\cos \alpha - \beta = \frac{3-4\sqrt{3}}{10}$

Q006 See the numbered solution; the three-angle challenge gives $\sin(\alpha+\beta+\gamma) = \frac{117}{125}$.

Q007 $\tan 105° = -2+\sqrt{3}$; $\theta = \frac{\pi}{4}$

Q008 The first pair gives $\theta = \frac{\pi}{4}$; the second pair also gives $\theta = \frac{\pi}{4}$.

Q009 See the numbered solution; the two slopes are $m = 2/11, -2$, with triangle areas $49/165$ and $25/3$.

Q010 $\sin 2\alpha = -\frac{4\sqrt{2}}{9}$, $\cos 2\alpha = \frac{7}{9}$, $\tan 2\alpha = -\frac{4\sqrt{2}}{7}$

Q011 $\sin 2\alpha = \frac{4\sqrt{5}}{9}$, $\cos 2\alpha = \frac{1}{9}$, $\tan 2\alpha = 4\sqrt{5}$

Q012 For the challenge, $\cos 3\theta = -\frac{117}{125}$, $\tan 2\theta = \frac{24}{7}$, so the product is $-\frac{2808}{875}$.

Q013 $\sin\frac{\alpha}{2} = \frac{1}{\sqrt{5}}$, $\cos\frac{\alpha}{2} = \frac{2}{\sqrt{5}}$, $\tan\frac{\alpha}{2} = \frac{1}{2}$, $\cos\frac{5\pi}{8} = -\frac{\sqrt{2-\sqrt{2}}}{2}$

Q014 $\sin\frac{\alpha}{2} = \sqrt{\frac{2}{3}}$, $\cos\frac{\alpha}{2} = \frac{1}{\sqrt{3}}$, $\tan\frac{\alpha}{2} = \sqrt{2}$, $\sin\frac{3\pi}{8} = \frac{\sqrt{2+\sqrt{2}}}{2}$

Q015 Use the signed half-angle formulas; for the challenge,

$\cos(A/2 - B/2) = \frac{\sqrt{3+\sqrt{5}} - \sqrt{15-5\sqrt{5}}}{6}$,

$\sin(A/4) = \sqrt{\frac{1-\sqrt{5/6}}{2}}$.

Q016 a $\theta = 0, \frac{2\pi}{3}, \pi, \frac{4\pi}{3}$; b $\frac{\pi}{3} \le \theta \le \frac{5\pi}{3}$

Q017 $\theta = \frac{\pi}{2}, \frac{7\pi}{6}, \frac{3\pi}{2}, \frac{11\pi}{6}$

Q018 See the numbered solution: replace double angles, solve in $s = \sin\theta$ or $c = \cos\theta$, and translate the accepted ranges.

Q019 $r = 2$, $\alpha = -\frac{\pi}{3}$, $\sin\theta - \sqrt{3}\cos\theta = 2\sin\theta - \frac{\pi}{3}$

Q020 $\sqrt{2}\sin\theta - \frac{\pi}{4}$, $\sqrt{2}\sin\theta + \frac{\pi}{4}$, $2\sqrt{3}\sin\theta - \frac{\pi}{6}$, $\sqrt{10}\sin\theta - \arctan\frac{1}{3}$

Q021 See the numbered solution; every result has $r = \sqrt{a^2+b^2} > 0$ and α chosen in $(-\pi, \pi)$.

Q022 a $\theta = \frac{\pi}{2}, \pi$; b $0 \le \theta < 2\pi$

Q023 $0 \le \theta \le \frac{\pi}{4}$ or $\frac{5\pi}{4} \le \theta < 2\pi$

Q024 See the numbered solution; synthesize first, then solve in the shifted variable over its full range.

Q025 First: $\max = 2, \min = -2$. Second: $\max = 7$ at $\theta = 0$; $\min = -2$ at $\theta = \frac{2\pi}{3}, \frac{4\pi}{3}$

Q026 For the first, $\max = 1, \min = -\frac{73}{48}$; for the second, use $t = \cos\theta$ and the endpoint/vertex test.

Q027 See the numbered solution: synthesize $a\sin\theta + b\cos\theta$ or substitute $t = \cos\theta$, then test the full allowed range.

Q028 B

Q029 C

Q030 B

Q031 A

Q032 B

Q033 A

Q034 B

Q035 A

Q036 B

Q037 B

Q038 A

Q039 B

Q040 A

Q041 A

Q042 B

Q043 A

Q044 A

Q045 B

Q046 B

Q047 A

Q048 A

Q049 A

Q050 B

Q051 B

Q052 B

Q053 A

Q054 B

Quiz MCQ 1 D

Quiz MCQ 2 A

Quiz MCQ 3 A

Quiz MCQ 4 B

Quiz WRITTEN 5 $\max = \frac{9}{4}, \min = 0$

Quiz WRITTEN 6 $-\frac{4\sqrt{2}}{9}, \frac{7}{9}, -\frac{4\sqrt{2}}{7}$

Quiz WRITTEN 7 $2\sqrt{3}\sin\theta - \frac{\pi}{6}$

Quiz WRITTEN 8 $[0, \frac{\pi}{4}] \cup [\frac{5\pi}{4}, 2\pi)$